Methodology

Developing software is difficult, and without a meaningful methodology it becomes both difficult and painful. Technology is solving newer and newer problems, and business models are being rewritten almost every three years. Remember Orkut was overshadowed by Hi5, and Hi5 was overshadowed by Facebook, as the social networking platform of choice, all within about five years. Such rapid changes require high velocity programming, which in turn, requires an agile method like Scrum or XP.

As software engineering continues to evolve, we are beginning to see the convergence of the 'development' and the 'operations' functions; so as a developer, you should be comfortable writing new code as well as fixing legacy ones even while answering questions on your support portal.

Browse through:

- Extreme Programming - *http://www.extremeprogramming.org/*
- Scrum - *https://www.scrum.org/*
- DevOps - *http://devops.com/*

Philosophy

It's true that programmers live and work adhering to a certain philosophy. You are likely to come across terms like the DRY Principle, Inversion of Control, etc. To make it easy for you to discover a work philosophy that suits your personality, it's highly recommended that you read the following essays to truly distinguish yourself.

Essays you must read:

- The Cathedral and the Bazaar, by Eric S Raymond - *http://www.catb.org/~esr/writings/cathedral-bazaar/cathedral-bazaar/*
- Beyond Code, by Rajesh Setty - *http://www.rajeshsetty.com/resources/books/beyond-code/*
- Mythical Manmonth, by Fredrick Brooks - *http://en.wikipedia.org/wiki/The_Mythical_Man-Month*

Copyright laws

As a programmer, and especially after you've read this article, you'll be tempted to try and reuse code as much as possible but this does come with some caveats. Usage of someone else's work requires their consent, and most of them advertise the terms of this consent by publishing their work under a set of copyright laws. So an awareness of the various licensing mechanisms like Creative Commons, Copy Left, BSD, etc, will ensure that you don't get on the wrong side of the law, and that you use the code according to the licence terms under which it has been released.

Business analysis

A project involves multiple stakeholders. You should learn the fine art of mapping business requirements to architectural diagrams and then translate these architectural diagrams to code. When this step is done right, fewer bugs reach the customer, and you spend less time dealing with a frustrated client. An understanding of the client's business is essential to achieve this.

Project management

Though you needn't know all the nuances of being a project manager, it is essential that you know how to prioritise tasks. You must know how these tasks are related to each other and how a delay in one critical task leads to the whole project being delayed. To be on top of the schedule you should be organised in your activities to ensure that everything works on the day of deployment. Choose the method that works for you. Here are a couple of books to get you started.

Browse through:

- Seven Habits of Highly Effective People, by Franklin Covey - *http://www.franklincovey.com/*
- Getting Things Done, by David Allen - *http://www.gtdtimes.com/*

Aesthetics

A thing of beauty is a joy for ever. Without an aesthetic appeal, a product is unlikely to find acceptance, no matter what features you put into it. Visual appeal in terms of colour, layout, typography, fonts and other eye candy is a must, and as a Web 2 programmer, this is an essential skill for your success.

Downloading code

There are several online repositories from where you can download code snippets or an entire project, and modify it to your heart's content (check the accompanying licence for restrictions).

The most popular among source code repositories are *https://github.com/* and *http://sourceforge.net/*. Check out the trending repositories on GitHub to see and participate in the most active projects. SourceForge has a 'hall of fame' called 'Project of the month'. Another notable mention is *http://freecode.com/* which was called 'Fresh Meat' a few years back. It has a list of the most popular projects. Google Code *(http://code.google.com/)* has several popular APIs and tools that you might find useful.

Keeping up with others by hanging out and networking

Most of the technologies mentioned above have well documented manuals and 'Getting started' guides. The forums on these sites are very active and responses to questions often get posted within a few seconds. There are several online sites that have presentation slides, speaker